

MATH 4162/Math 6201
Numerical methods for differential equations
Fall 2025

Homework problem set #2

Show all necessary steps of working in the given space

Name: _____

ID: _____

Upload the assignment solution no later than

23:59 on Oct 24, 2025

Full marks 50

1. [25pts]

(a) For the advection-diffusion equation

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}, \quad a > 0 \quad \nu > 0$$

write the numerical scheme using upwind for advection and center for diffusion.

(b) Show that the exact amplification factor

$$G = \frac{u(x, t + \Delta t)}{u(x, t)}$$

satisfies $|G| = 1$ for the Fourier mode

$$u(x, t) = e^{i(kx + \omega t)}.$$

(c) Using von Neumann method, show that amplification factor of the scheme is

$$G = 1 - (C + 2D)(1 - \cos \beta) - Ci \sin \beta$$

where

$$C = \frac{a\Delta t}{\Delta x}, \quad D = \frac{\nu\Delta t}{\Delta x^2}, \quad \beta = k\Delta x.$$

(d) Plot the stability curve $G(\beta)$ by taking $C + 2D = 1$ and $C = 1$ for $\beta \in [-\pi, \pi]$.

Based on this plot, if the maximum wavenumber associated with the Fourier mode is $k_{\max} = 1000$, what should be the value of Δx .

Choose $C + 2D = 1/2$ and $C = 1$. Justify if the scheme will be stable for this choice of the parameters.

2. [25pts]

- (a) Using Taylor series expansion, find the leading order error terms for the given scheme (in Q1).
(b) Find the dispersion relation of the modified equation:

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = \underbrace{\left[\nu + \frac{a\Delta x}{2} - \frac{a^2\Delta t}{2} \right]}_{\nu_{eff}} \frac{\partial^2 u}{\partial x^2} + \underbrace{\left[\frac{-a\Delta x^2}{6} + a\nu\Delta t \right]}_{\phi_{eff}} \frac{\partial^3 u}{\partial x^3}$$

(which is given for equation and scheme in Q1).

- (c) Find the decay rate $e^{-\text{Im}\omega(k)t}$ of the modified equation and real part of the phase speed.
(d) Set $\phi_{eff} = 0$, determine the CFL number for the scheme to be stable?
Plot the stability curve $G(\theta)$ under this condition.
Show that $|G(\theta)| = 1$ requires $\Delta x = (4\nu)/a$.
(e) Select $\phi_{eff} = 0$. Using dispersion relation, show that this choice can eliminate the dispersion error; however, show that the dissipation error of numerical scheme is given by the rate $e^{-(2/3)\nu k^2 t}$